# PRACTICAL NO.1

## Q.1) Program to display name and address.

**PROGRAM: -**

```
name = input("Enter your name: ")
address = input("Enter your address: ")
print("Your name is:", name)
print("You live in:", address)
```

**OUTPUT: -**

```
Enter your name: Vikrant Sutar
Enter your address: Sutar malla, Atpadi
Your name is: Vikrant Sutar
You live in: Sutar malla, Atpadi
```

# PRACTICAL NO.2

**Q.2) Program to accept two numbers and display addition, subtraction, multiplication, division and modules.**

**PROGRAM: -**

```
def calculator(a, b):
    print("Addition:", a + b)
    print("Subtraction:", a - b)
    print("Multiplication:", a * b)
    print("Division:", a / b)
    print("Modulus:", a % b)
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
calculator(a, b)
```

**OUTPUT: -**

```
Enter first number: 100
Enter second number: 50
Addition: 150
Subtraction: 50
Multiplication: 5000
Division: 2.0
Modulus: 0
```

# PRACTICAL NO.3

**Q.3) Program to calculate Factorial of a given number.**

**PROGRAM: -**

```
def factorial():
    num = int(input("Enter number: "))
    fact = 1
    for i in range(1, num + 1):
            fact *= i
    print("Factorial:", fact)
factorial()
```

**OUTPUT: -**

```
Enter number: 20
Factorial: 2432902008176640000
```

# PRACTICAL NO.4

**Q.4) Program to create a list of 100 numbers and separate those numbers in two different lists one includes odd numbers other includes even numbers.**

**PROGRAM: -**

```
def even_odd():
    numbers = list(range(1, 101))
    even = []
    odd = []
    for num in numbers:
            if num % 2 == 0:
                        even.append(num)
            else:
                        odd.append(num)
    print("Even numbers:", even)
    print("Odd numbers:", odd)
even_odd()
```

**OUTPUT: -**

```
Even numbers: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24,
26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54,
56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84,
86, 88, 90, 92, 94, 96, 98, 100]
Odd numbers: [1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25
, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49, 51, 53, 5
5, 57, 59, 61, 63, 65, 67, 69, 71, 73, 75, 77, 79, 81, 83,
85, 87, 89, 91, 93, 95, 97, 99]
```

# PRACTICAL NO. 5

**Q.5) Program to display maximum and minimum number from the given list.**

**PROGRAM: -**

```
def max_min(lst):
    max_num = lst[0]
    min_num = lst[0]
    for num in lst:
        if num > max_num:
            max_num = num
        elif num < min_num:
            min_num = num
    return max_num, min_num
lst1 = [13,3,9,88,15,12]
maximum, minimum = max_min(lst1)
print("Maximum is:", maximum)
print("Minimum is:", minimum)
```

**OUTPUT: -**

```
Maximum is: 88
Minimum is: 3
```

# PRACTICAL NO. 6

Q.6) **Program to demonstrate slicing.**

**PROGRAM: -**

```
def slicing():
lst = [1,20,30,40,50,60,70]
    print("Original List:", lst)
    print("First 4 elements:", lst[:4])
    print("Last 2 elements:", lst[-2:])
    print("Reverse:", lst[::-1])
    print("Middle elements (index 3 to 6):", lst[3:6])
slicing()
```

**OUTPUT: -**

```
Original List: [1, 20, 30, 40, 50, 60, 70]
First 4 elements: [1, 20, 30, 40]
Last 2 elements: [60, 70]
Reverse: [70, 60, 50, 40, 30, 20, 1]
Middle elements (index 3 to 6): [40, 50, 60]
```

# PRACTICAL NO.7

**Q.7) Program to demonstrate set operators (union , intersection, minus).**

**PROGRAM: -**

```
def set_oper():
    set1 = {2,4,6,8}
    set2 = {1,3,5,7}
    print("Set 1:", set1)
    print("Set 2:", set2)
    print("Union:", set1 | set2)
    print("Intersection:", set1 & set2)
    print("Difference (set1 - set2):", set1 - set2)
    print("Difference (set2 - set1):", set2 - set1)
set_oper()
```

**OUTPUT: -**

```
Set 1: {8, 2, 4, 6}
Set 2: {1, 3, 5, 7}
Union: {1, 2, 3, 4, 5, 6, 7, 8}
Intersection: set()
Difference (set1 - set2): {8, 2, 4, 6}
Difference (set2 - set1): {1, 3, 5, 7}
```

# PRACTICAL NO.8

**Q.8) Program to print current date and time.**

**PROGRAM: -**

```
from datetime import datetime
current = datetime.now()
print("Current Date and Time:", current)
```

**OUTPUT: -**

```
Current Date and Time: 2026-03-31 10:46:45.337126
```

# PRACTICAL NO. 9

**Q.9) Program to Today's Year, Month, and Date.**

**PROGRAM: -**

```
from datetime import date
cur_date = date.today()
print("Year is:", cur_date.year)
print("Month is:", cur_date.month)
print("Day is:", cur_date.day)
```

**OUTPUT: -**

```
Year is: 2026
Month is: 3
Day is: 31
```

# PRACTICAL NO.10

## Q.10 ) Program to convert Date to String.

**PROGRAM: -**

```
from datetime import datetime
current = datetime.now()
datestr = current.strftime("%Y-%m-%d %H:%M:%S")
print("Date as String:", datestr)
print(type(datestr))
```

**OUTPUT: -**

```
Date as String: 2026-03-31 11:09:15
<class 'str'>
```

# PRACTICAL NO.11

**Q.11) Program to display the Calendar of a given month.**

**PROGRAM: -**

```
import calendar

year = int(input("Enter year of your choice: "))
month = int(input("Enter month of your choice: "))

print(calendar.month(year, month))
```

**OUTPUT: -**

```
Enter year of your choice: 2005
Enter month of your choice: 4
     April 2005
Mo Tu We Th Fr Sa Su
                1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30
```

# PRACTICAL NO.12

## Q.12) Program to display calendar of the given year.

**PROGRAM: -**

```
import calendar
year = int(input("Enter year: "))
print(calendar.calendar(year))
```

**OUTPUT: -**

```
Enter year: 2005
                                  2005

      January                   February                   March
Mo Tu We Th Fr Sa Su      Mo Tu We Th Fr Sa Su      Mo Tu We Th Fr Sa Su
                   1  2          1  2  3  4  5  6          1  2  3  4  5  6
 3  4  5  6  7  8  9       7  8  9 10 11 12 13       7  8  9 10 11 12 13
10 11 12 13 14 15 16      14 15 16 17 18 19 20      14 15 16 17 18 19 20
17 18 19 20 21 22 23      21 22 23 24 25 26 27      21 22 23 24 25 26 27
24 25 26 27 28 29 30      28                        28 29 30 31
31

       April                     May                       June
Mo Tu We Th Fr Sa Su      Mo Tu We Th Fr Sa Su      Mo Tu We Th Fr Sa Su
              1  2  3                            1          1  2  3  4  5
 4  5  6  7  8  9 10       2  3  4  5  6  7  8       6  7  8  9 10 11 12
11 12 13 14 15 16 17       9 10 11 12 13 14 15      13 14 15 16 17 18 19
18 19 20 21 22 23 24      16 17 18 19 20 21 22      20 21 22 23 24 25 26
25 26 27 28 29 30         23 24 25 26 27 28 29      27 28 29 30
                          30 31

        July                    August                  September
Mo Tu We Th Fr Sa Su      Mo Tu We Th Fr Sa Su      Mo Tu We Th Fr Sa Su
              1  2  3       1  2  3  4  5  6  7                 1  2  3  4
 4  5  6  7  8  9 10       8  9 10 11 12 13 14       5  6  7  8  9 10 11
11 12 13 14 15 16 17      15 16 17 18 19 20 21      12 13 14 15 16 17 18
18 19 20 21 22 23 24      22 23 24 25 26 27 28      19 20 21 22 23 24 25
25 26 27 28 29 30 31      29 30 31                  26 27 28 29 30

      October                   November                  December
Mo Tu We Th Fr Sa Su      Mo Tu We Th Fr Sa Su      Mo Tu We Th Fr Sa Su
                   1  2          1  2  3  4  5  6                 1  2  3  4
 3  4  5  6  7  8  9       7  8  9 10 11 12 13       5  6  7  8  9 10 11
10 11 12 13 14 15 16      14 15 16 17 18 19 20      12 13 14 15 16 17 18
17 18 19 20 21 22 23      21 22 23 24 25 26 27      19 20 21 22 23 24 25
24 25 26 27 28 29 30      28 29 30                  26 27 28 29 30 31
31
```

# PRACTICAL NO.13

**Q.13) Program to demonstrate File input.**

**PROGRAM: -**

```
file = open("sample.txt", "r")
print("file is open")
content = file.read()
print(content)
file.close()
```

**OUTPUT: -**

```
file is open
My name is vikrant Vinayak sutar..
This is demonstration of python file input..
```

# PRACTICAL NO.14

## Q.14) Program to demonstrate file output.

**PROGRAM: -**

```
file = open("sample.txt", "a")
file.write("Hello, this is file output example.")
file.write("Welcome to python coding")
file.close()
print("Data written successfully.")
```

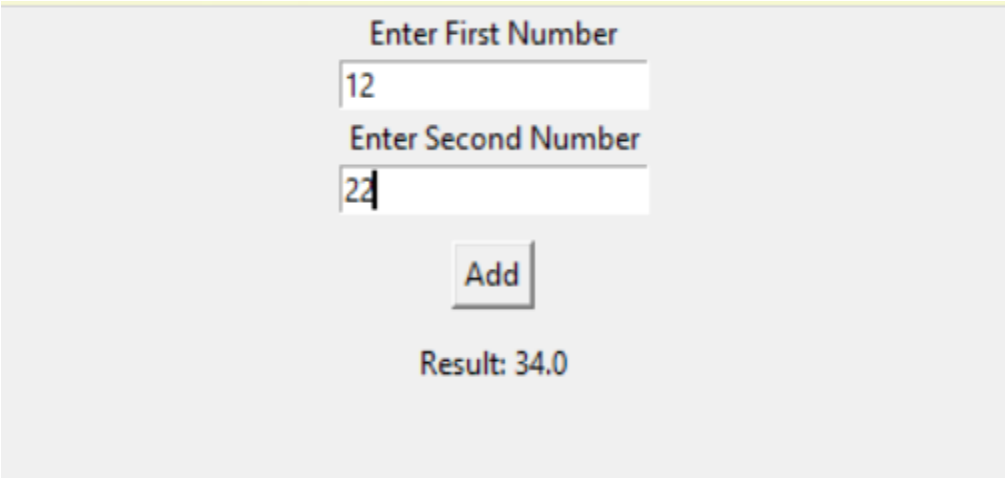
**OUTPUT: -**

```
Data written successfully.
```

# PRACTICAL NO.15

## Q.15) Program two add two numbers using GUI.

**PROGRAM: -**

```
import tkinter as tk
def add_numbers():
    try:
        num1 = float(entry1.get())
        num2 = float(entry2.get())
        result = num1 + num2
        result_label.config(text="Result: " + str(result))
    except:
        result_label.config(text="Invalid Input")
root = tk.Tk()
root.title("Addition Calculator")
root.geometry("300x200")
tk.Label(root, text="Enter First Number").pack()
entry1 = tk.Entry(root)
tk.Label(root, text="Enter Second Number").pack()
entry2 = tk.Entry(root)
entry2.pack()
tk.Button(root, text="Add", command=add_numbers).pack(pady=10)
result_label = tk.Label(root, text="Result: ")
result_label.pack()
root.mainloop()
```

**OUTPUT: -**